

2.31 $y := v$

$$\Rightarrow \begin{cases} -\frac{d^2 y(x)}{dx^2} = f & 0 < x < 1 \\ v(0) = v(1) = 0 \end{cases}$$

$$x_i = 0 + i h, \quad i = 0, \dots, N-1$$

$$x_i = i h$$

$$h = \frac{1}{N-1}, \quad y_i \approx y(x_i)$$

$$y_0 = 0$$

$$y_{N-1} = 0$$

$$-\frac{d^2 y(x)}{dx^2} = -y''(x)$$

$$\stackrel{(30)}{\Rightarrow} -\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} = f(x_i), \quad \forall i = \overline{1, N-2}$$

$$\Rightarrow -(y_{i-1} - 2y_i + y_{i+1}) = h^2 \cdot f(x_i), \quad \forall i = \overline{1, N-2}$$

$$\text{pt } i=1 \Rightarrow -(y_0 - 2y_1 + y_2) = h^2 f(x_1)$$

$$\Rightarrow 2y_1 - y_2 = h^2 f(x_1)$$

$$\Rightarrow -(y_{i-1} + 2y_i - y_{i+1}) = h^2 f(x_i), \quad \forall i = \overline{2, N-3}$$

$$\text{pt } i=N-2 \Rightarrow -y_{N-3} + 2y_{N-2} = h^2 f(x_{N-2})$$

$$\Rightarrow A = \begin{bmatrix} 2 & -1 & 0 & 0 & & \\ -1 & 2 & -1 & 0 & & \\ 0 & -1 & 2 & -1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{bmatrix}$$

$$b = h^2 \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_{N-2}) \end{bmatrix}$$

$$\Rightarrow A \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_{N-2} \end{bmatrix} = b$$

(a) $A = \begin{bmatrix} 2 & -1 & 0 & 0 & & \\ -1 & 2 & -1 & 0 & & \\ 0 & -1 & 2 & -1 & & \\ & & \ddots & \ddots & \ddots & \\ 0 & & & -1 & 2 & -1 \\ & & & & -1 & 2 \end{bmatrix}$, A este tridiagonal

$m = N-2$

$A^T = A \Rightarrow A$ este simetrică

Fie $x \in \mathbb{R}^n \setminus \{0\} \Rightarrow \exists x_i \neq 0$

$x^T A x = [2x_1 - x_2, -x_1 + 2x_2 - x_3, -x_2 + 2x_3 - x_4, \dots$

$\dots, -x_{m-2} + 2x_{m-1} - x_m, -x_{m-1} + 2x_m] \cdot \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} =$

$= 2x_1^2 - x_1x_2 - x_1x_2 + 2x_2^2 - x_2x_3 - x_2x_3 + 2x_3^2 - x_3x_4 + \dots$

$\dots - x_{m-1}x_{m-2} + 2x_{m-1}^2 - x_{m-1}x_m - x_{m-1}x_m + 2x_m^2 =$

$= x_1^2 + (x_1^2 - 2x_1x_2 + x_2^2) + (x_2^2 - 2x_2x_3 + x_3^2) + \dots$

$+ (x_{m-1}^2 - 2x_{m-1}x_m + x_m^2) + x_m^2 = \underbrace{x_1^2}_{\geq 0} + \underbrace{(x_1 - x_2)^2}_{\geq 0} + \underbrace{(x_2 - x_3)^2}_{\geq 0} + \dots$

$+ \underbrace{(x_{m-1} - x_m)^2}_{\geq 0} + \underbrace{x_m^2}_{\geq 0} \geq 0 \iff ((x_i - x_{i+1})^2 \geq 0)$

$\Rightarrow A$ este pozitiv definită

(b) $\Rightarrow A$ este tridiagonală SPD $\Rightarrow$ Putem să aplicăm algoritmul de la 2.29